A Single Drone. Forty Sensors. Thirty-Five Minutes.

Energy-Efficient UAV Path Planning in IoT Networks: A Deep Reinforcement Learning Aided Approach

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The scene

A remote sensor field.

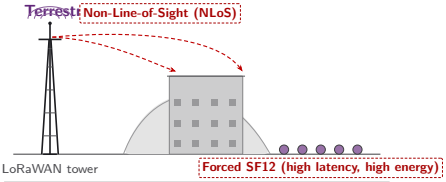
Forty sensors.

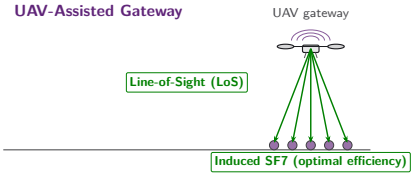
One drone.

35 minutes of battery.

The drone is the only gateway. Every uncollected sensor is lost data.

And it is also a protocol-controller, not just a data-mule.





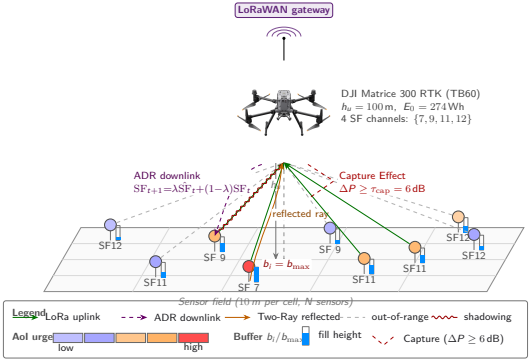
Two coupled problems, one control input

Physical

Limited battery, large grid, partial coverage. Drone position determines link budget.

Protocol: *mobility-blind ADR*

LoRaWAN's Adaptive Data Rate is engineered for static gateways. Under a moving UAV it triggers the **SF monoculture**: every sensor converges on SF12, intra-SF collisions collapse throughput.



UAV position is the shared control input.

Gap: no prior work addresses physical *and* protocol jointly on a mobile gateway.

What prior work does – and misses

MILP Zorbas et al.	Optimal SF allocation. Exponential in N; intractable $N>20$. Assumes static channel conditions.
FlyingLoRa Xiong et al.	Offline 3D trajectory + scheduling. No ADR dynamics, no Capture Effect. Sensor rates treated as fixed.
DRL, flat MLP Liu+, Challita+	Online path learning. State $[\mathrm{uav}, s_1, \ldots, s_N]$ bakes in sensor order. Retrain needed per N.

Gap 1: protocol physics

No prior work couples trajectory control with EMA-ADR, Capture Effect, and duty cycles on a **moving gateway**.

Gap 2: representation

No prior work applies a **permutation-invariant** set-structured policy, or measures what the flat-MLP index-sensitivity costs at scale.

Gap 2 is the dominant bottleneck.
That is the project's main discovery.

Five things in 15 minutes

1. **The problem** two coupled constraints, no closed form
2. **My approach** DQN trained across 16 conditions simultaneously
3. **What I found** performance collapses at scale; a greedy heuristic beats it
4. **Why, and the fix** representational ceiling diagnosed; permutation-invariant architecture
5. **The evidence** NDR $+333\%$, energy $\times 12$, all $p < 10^{-4}$, 50 seeds

Problem formulation

Objective: maximise total data collected

$$\max_{\pi} \sum_{t=0}^{T} \sum_{i=1}^{N} D_{i,t}$$

Subject to:

$$\mathrm{RSSI}_{i,t}(p_t) \geq \mathrm{RSSI}_{\min}(SF_{i,t}) \qquad \forall i$$

$$E_{t+1} = E_t - P(a_t)\Delta t \geq 0 \qquad \forall t$$

$$\textstyle\sum_{i \in \mathcal{N}} \mathbf{1}[SF_{i,t}{=}k] \leq 1 \qquad \forall k \in \{7, \ldots, 12\}$$

$$p_{t+1} = p_t + v(a_t)\,\Delta t$$

$D_{i,t}$: bytes from sensor i at step t. π: control policy. E_t: battery. $SF_{i,t}$: spreading factor.

Three forces act simultaneously:

Battery

$$E_{t+1} = E_t - P(a_t)\Delta t$$

$P_{\text{hover}} > P_{\text{flight}}$; hovering costs more.

Coverage

$$P_r \propto d^{-4}$$

Two-Ray: halve distance $\Rightarrow$ 16× power.

Protocol lag

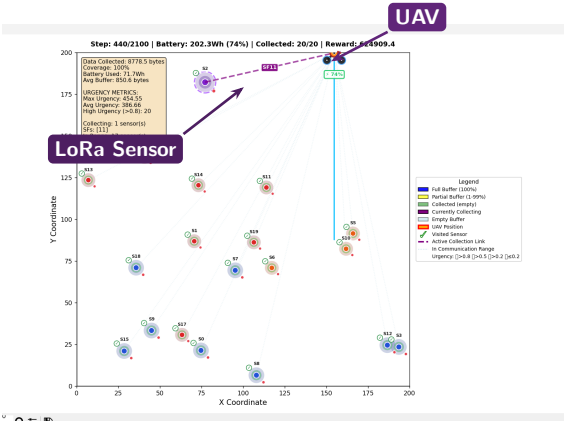
$$\bar{R}_t = \lambda R_t + (1-\lambda)\bar{R}_{t-1}$$

EMA-ADR, $\lambda{=}0.1$; SF lags $\sim$10 s.

Non-convex. Non-stationary. No closed form.

$\Rightarrow$ trained a **Deep Q-Network** to find π.

Experimental setup: the simulator



Environment at a glance

- **Grid**: 200×200 m, UAV at 100 m altitude
- **LoRa sensors (S0–S19)**: ring colour = buffer fill (blue = full, green = empty)
- **UAV** (orange): SF11 link; dashed line = active link
- **Urgency rings**: AoI priority — darker ⇒ data loss imminent
- **Dotted circle**: comm. range — 17 of 20 reachable
- **Episode**: 2100 steps ≈ 35 min

The agent: DQN with domain randomisation

Observation (4-frame stack)

UAV position · battery · per-sensor {buffer fill, urgency, link quality}

Domain randomisation

4 grid sizes $\times$ 4 sensor counts $= 16$ conditions simultaneously. One policy, deployed everywhere.

5 discrete actions

N · S · E · W · Hover, chosen every second. Flat MLP $[512, 512, 256]$, ReLU, Adam.

At small scale: parity

Three comparators:

Nearest Greedy	Move to closest buffered sensor. No protocol knowledge.
SF-Aware Greedy	Adds protocol awareness; picks highest-throughput sensor.
TSP Oracle	Optimal offline route. Routing upper bound.

50 seeds per condition, 5 grid sizes $(100\times100, N{=}10) \rightarrow (500\times500, N{=}50)$. Welch's t-test.

$$100\%$$

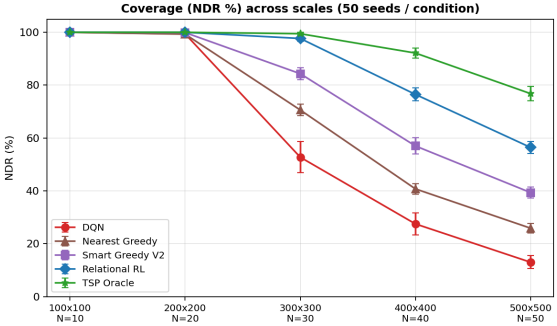
DQN NDR at $(100\times100,\ N{=}10)$

$$99.4\%$$

DQN NDR at $(200\times200,\ N{=}20)$

All agents at parity. The DQN is working.

At scale: collapse



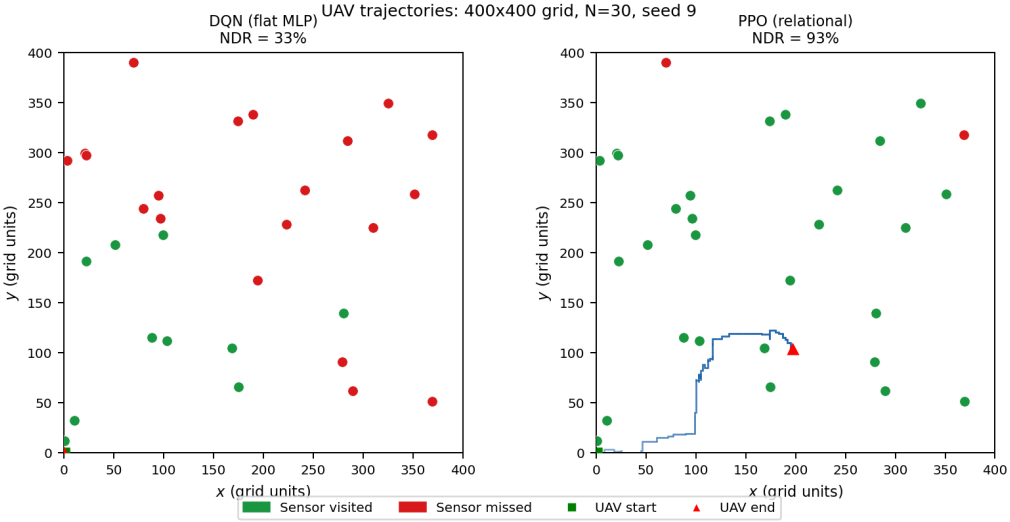
Agent	$(300, 30)$	$(500, 50)$
DQN	52.7%	13.0%
Nearest Greedy	70.6%	25.9%
SF-Aware Greedy	84.3%	39.4%

13.0%

DQN NDR at $(500 \times 500,\ N{=}50)$

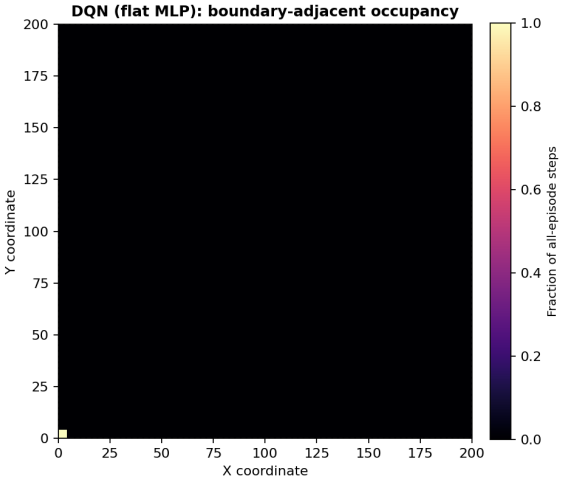
Both greedy baselines beat the protocol-aware DQN.

What is the DQN actually doing?



Same 30 sensors. Same starting position.

Diagnostic: 100% boundary-cell occupancy



20 deterministic episodes,
$(200 \times 200, N{=}20)$

$$100\%$$

boundary-cell occupancy

Median 0 wall collisions:
boundary avoidance *was* learned.

But the interior is never entered.

Reward penalises both boundary hits and starvation. The agent still patrols the perimeter.

Physically capable. Architecturally constrained.

The cause: representational ceiling

Not reward. Not training. Not physics.

The flat MLP consumes $[\mathrm{uav},\ s_1,\ s_2,\ \ldots,\ s_N]$ as a single concatenated vector.

The problem

Sensor order is part of the representation. Swapping sensor 1 and sensor 2 $\Rightarrow$ the policy sees a completely different input.

At small scales: memorise the fixed layout. At large scales: cannot generalise.

The consequence

Perimeter patrol is the cheapest *stable* trajectory the network can express under this encoding.

The architecture cannot treat the sensor set as a set.

The fix: an architectural change

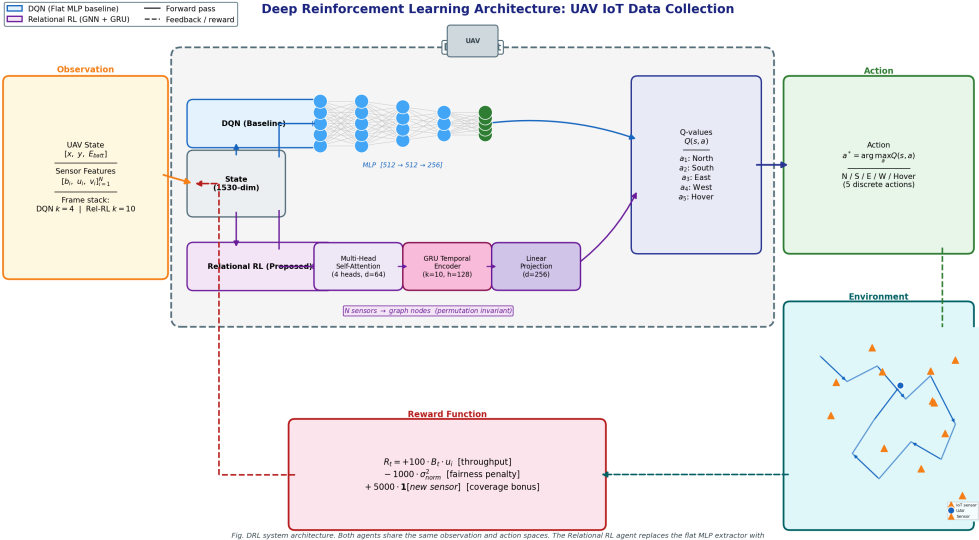
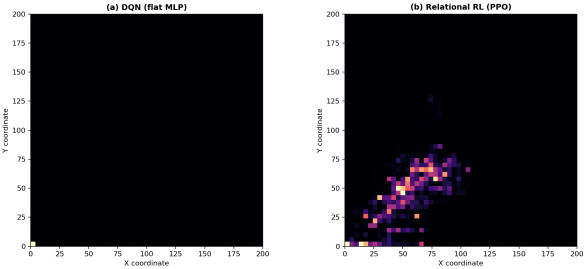


Fig. DRL system architecture. Both agents share the same observation and action spaces. The Relational RL agent replaces the flat MLP extractor with sensor self-attention (spatial GNN) and a GRU temporal encoder, enabling permutation-invariant reasoning over a variable number of IoT sensors.

Grid: 100–1000 units | N = 10–40 sensors

Diagnostic resolved



20 seeds, 200×200, $N=20$

Metric	DQN	Rel. RL
Edge occupancy	100%	$\mathbf{21.5\%}$
Wall hits (mean)	21.8	$\mathbf{0}$
Full coverage	$15/20$	$\mathbf{20/20}$

Left: DQN trapped on the perimeter.
Right: Relational RL covers the grid.

Root cause confirmed. Fix validated.

Is it the architecture or the algorithm?

Controlled experiment: swap one factor at a time.

	DQN (flat MLP)	PPO (flat MLP)	PPO (relational)
300×300, $N=30$	52.7%	58.9%	$\mathbf{96.6}\%$
400×400, $N=40$	27.5%	42.5%	$\mathbf{82.2}\%$
500×500, $N=50$	13.0%	24.7%	$\mathbf{58.4}\%$

Algorithm effect (DQN $\rightarrow$ PPO)

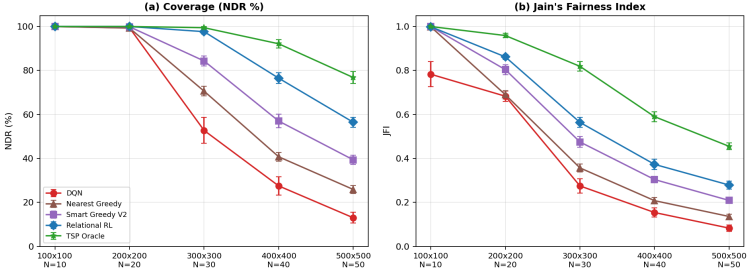
$+6$ to $+15$ pp NDR at $N\geq30$

Architecture effect (flat $\rightarrow$ relational)

$+34$ to $+40$ pp NDR at $N\geq30$

Architecture effect is $3\times$ the algorithm effect. The gain is in the encoder, not the training recipe.

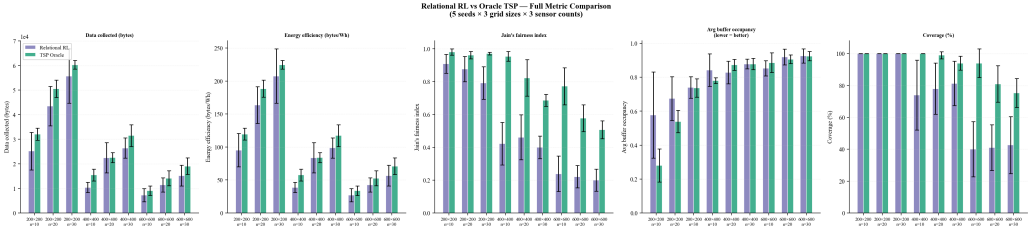
Results: scalability restored



Agent at $(500\times500, N{=}50)$	NDR	Jain's	Energy (B/Wh)
DQN (flat MLP)	13.0%	0.08	9.96
Nearest Greedy	25.9%	0.14	—
SF-Aware Greedy	39.4%	0.21	—
Relational RL	**56.4%**	**0.28**	**121.9**
TSP Oracle (upper bound)	76.8%	0.46	—

$+333\%$ **NDR** vs DQN. $+17$ **pp** vs best heuristic. Energy: $10.0 \rightarrow$ **121.9** B/Wh **($\times12$ vs DQN, $\times2.3$ vs PPO+FlatMLP)**

Results: how close to optimal?



TSP Oracle = omniscient upper bound (pre-computed tour, no mid-flight adaptation). 5 seeds × 3 grid sizes × 3 sensor counts. Error bars = std.

Throughput & efficiency

-14% overall; gap narrows at trained-scale grids (200×200).

Coverage

73% vs 94% NDR — driven by 600×600 OOD degradation.

Fairness

Oracle's uniform tour $\neq$ smarter.
Rel. RL weights **urgent** sensors.

Contributions and what's next

Contributions

1. Open Gymnasium env: Two-Ray + ADR + Capture Effect + shadowing, all coupled
2. Domain-randomised competence-gated curriculum
3. 50-seed benchmark against Greedy and TSP Oracle
4. **Architectural diagnostic:** flat-MLP perimeter heuristic as a representational ceiling
5. **Permutation-invariant fix:** pathology resolved, scalability curve shifted by $+333\%$ NDR

The diagnostic-and-fix pairing is the principal claim.

Future directions

- **Attention pooling** over sensor set (close the 20-point gap to TSP Oracle)
- **Multi-UAV** coordination (QMIX, MAPPO) (beyond single-drone battery envelope)
- **Hardware-in-the-loop** validation (sim-to-real robustness already tested)

Thank you.

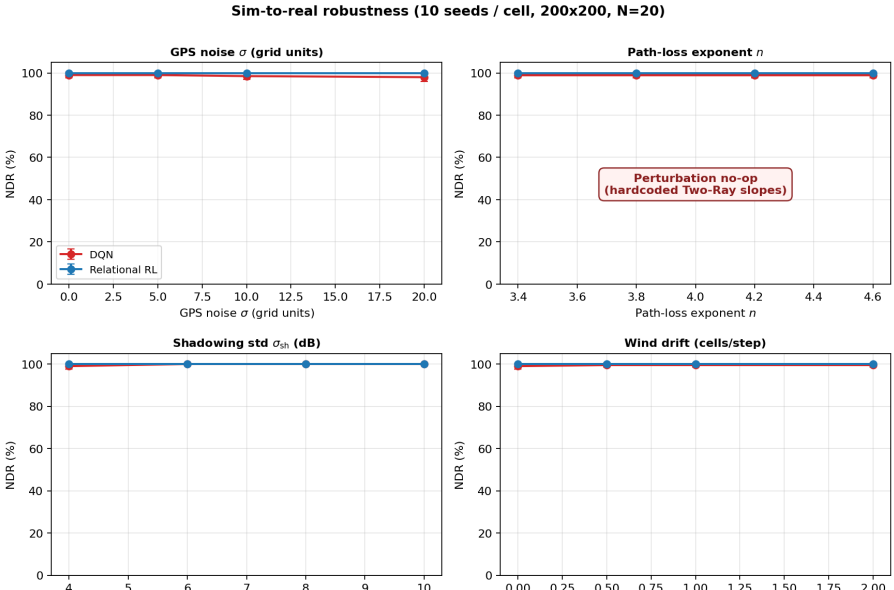
Questions?

Backup: ablation study

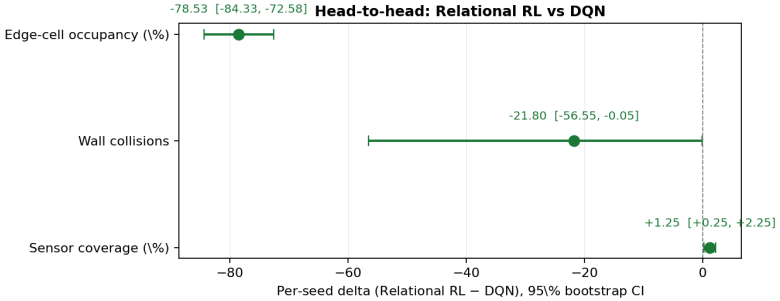
Metric	Full mean	Ablation mean	Cohen's d	p
Cumulative reward ($\times 10^5$)	8.24	2.88	+1.43	0.00012
NDR	0.186	0.138	+0.54	0.098
Jain's fairness	0.288	0.177	+1.23	0.00052
Gini (lower is better)	0.730	0.833	−1.32	0.00023
Energy efficiency (B/Wh)	51.4	35.8	+0.63	0.055

Ablation = no domain randomisation (A4). 20 seeds, Welch's t-test. Every large-effect component ($|d|>0.8$) reaches $p<0.001$.

Backup: sim-to-real robustness

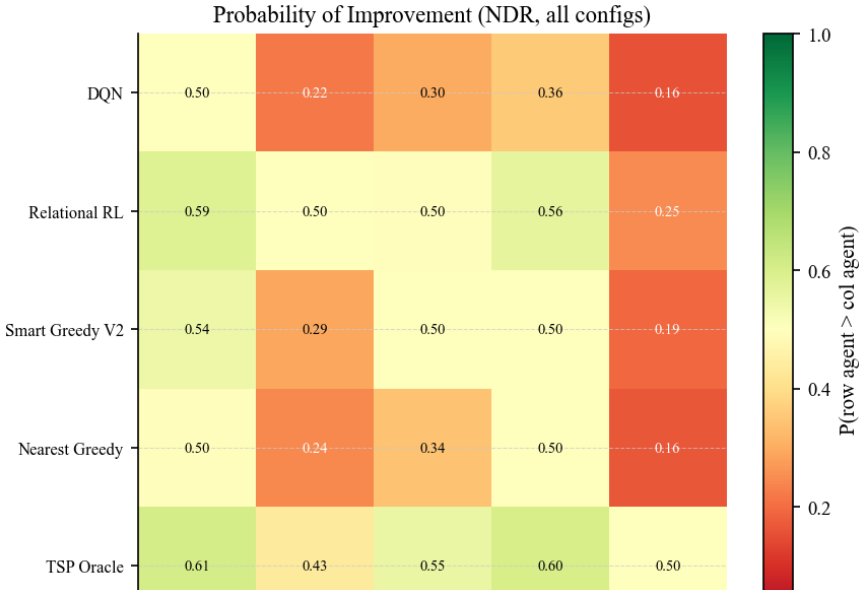


Backup: head-to-head forest plot

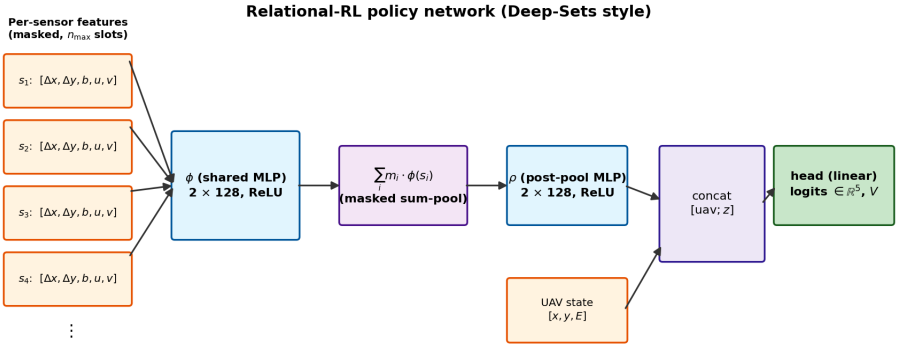


Seed-matched deltas (Relational RL − DQN) across three diagnostic metrics, 95% bootstrap CIs (n=10,000). All intervals exclude zero.

Backup: probability of improvement (50 seeds)



Backup: Relational RL network detail



Permutation invariance: reordering $\{s_i\}$ leaves the masked sum unchanged.

$$z = \rho\left(\sum_{i=1}^{n_{\max}} m_i\, \phi(s_i)\right)$$

ϕ: shared MLP per sensor Σ: masked sum-pool ρ: post-pool MLP